



## 2. Performance Benchmark

### 2a. Isolation Forest: Anomaly Detection

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 92.39%  | Share of actual attacks successfully flagged         |
| F1-Score                  | 96.04%  | Balanced measure combining Precision and Recall      |
| Accuracy                  | 97.93%  | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 7.61%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 92.39%  | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 3460           | FN 285            |
| Actual: Normal | FP 0              | TN 10026          |

### 2b. Random Forest: Attack Classification

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 100.00% | Share of actual attacks successfully flagged         |
| F1-Score                  | 100.00% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 100.00% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 0.00%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 100.00% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 0              | 0               | 0              |
| Act: ICMP | 0              | 0               | 0              |
| Act: UDP  | 0              | 0               | 3460           |

### 2c. Hybrid Cascaded

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 92.39%  | Share of actual attacks successfully flagged         |
| F1-Score                  | 96.04%  | Balanced measure combining Precision and Recall      |
| Accuracy                  | 97.93%  | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 7.61%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 92.39%  | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 3460           | FN 285            |
| Actual: Normal | FP 0              | TN 10026          |

2d. Response Latency

| Metric                   | Value    | Description                                                             |
|--------------------------|----------|-------------------------------------------------------------------------|
| Detection Time           | 344.4 ms | Average time taken to detect an attack                                  |
| Mitigation Response Time | 0.0 ms   | Interval between the anomaly flag and the FlowMod blocking rule install |

2e. Controller Resource Overhead

| Metric                         | Value | Description                                                           |
|--------------------------------|-------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | 1.34% | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | 1.57% | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | 2.68% | Ryu controller CPU usage during simultaneous detection and mitigation |

3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.20 | 2        | 2            | 2026-09-16 12:40:56 | 2026-09-16 12:42:07 |
| 10.0.0.21 | 2        | 2            | 2026-09-16 12:40:57 | 2026-09-16 12:42:08 |
| 10.0.0.22 | 2        | 2            | 2026-09-16 12:40:57 | 2026-09-16 12:42:08 |

#### 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector    | Confidence | Priority | Action        |
|---------------------|-----------|-------|-----------|------------|----------|---------------|
| 2026-09-16 12:40:56 | 10.0.0.20 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-16 12:40:57 | 10.0.0.21 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-16 12:40:57 | 10.0.0.22 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-16 12:41:07 | 10.0.0.20 | DDoS  | UDP Flood | 75.6%      | High     | Time Ban (1m) |
| 2026-09-16 12:41:07 | 10.0.0.20 | DDoS  | UDP Flood | 75.6%      | High     | Time Ban      |
| 2026-09-16 12:41:08 | 10.0.0.21 | DDoS  | UDP Flood | 77.6%      | High     | Time Ban (1m) |
| 2026-09-16 12:41:08 | 10.0.0.22 | DDoS  | UDP Flood | 76.3%      | High     | Time Ban (1m) |
| 2026-09-16 12:41:08 | 10.0.0.21 | DDoS  | UDP Flood | 77.6%      | High     | Time Ban      |
| 2026-09-16 12:41:08 | 10.0.0.22 | DDoS  | UDP Flood | 76.3%      | High     | Time Ban      |
| 2026-09-16 12:42:07 | 10.0.0.20 | DDoS  | UDP Flood | 77.6%      | High     | Time Ban (2m) |
| 2026-09-16 12:42:08 | 10.0.0.21 | DDoS  | UDP Flood | 87.1%      | High     | Time Ban (2m) |
| 2026-09-16 12:42:08 | 10.0.0.22 | DDoS  | UDP Flood | 83.8%      | High     | Time Ban (2m) |

#### 5. IP Attack History (Completed Sessions)

| Source IP | Vector    | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|-----------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.22 | UDP Flood | 0.6337   | 83.8% | High     | Phase 2 | 2        | 1m 11s   | 2026-09-16 12:42:08 | Attack persisted - escalating to time ban |
| 10.0.0.21 | UDP Flood | 0.6392   | 87.1% | High     | Phase 2 | 2        | 1m 11s   | 2026-09-16 12:42:08 | Attack persisted - escalating to time ban |
| 10.0.0.20 | UDP Flood | 0.5951   | 77.6% | High     | Phase 2 | 2        | 1m 11s   | 2026-09-16 12:42:07 | Attack persisted - escalating to time ban |
| 10.0.0.22 | UDP Flood | 0.6395   | 76.3% | High     | Phase 2 | 1        | 11s      | 2026-09-16 12:41:08 | Attack persisted - escalating to time ban |
| 10.0.0.21 | UDP Flood | 0.6464   | 77.6% | High     | Phase 2 | 1        | 11s      | 2026-09-16 12:41:08 | Attack persisted - escalating to time ban |
| 10.0.0.20 | UDP Flood | 0.6330   | 75.6% | High     | Phase 2 | 1        | 11s      | 2026-09-16 12:41:07 | Attack persisted - escalating to time ban |

#### Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

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SIGNATURE OVER PRINTED NAME

**Network Administrator**

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DATE